VTPSA Website: Technical Handover Notes

Part 1: Architecture and Initial Setup

12 November 2026

1 Overview

The VTPSA website is a static website that publishes results logs, club details, documents (constitution, rulebooks, minutes, reports) and photo galleries. All content lives as files in a Git repository. There is no database and no server-side code. The site is rebuilt and published automatically whenever changes are pushed to the `main` branch.

- **Live site:** `https://vtpsa.github.io`
- **Repository:** `https://github.com/VTPSA/vtpsa.github.io`

2 Design decisions and rationale

Static site instead of a database-backed web application.
The content volume is small (about 20 documents, 20 photos and a few result tables per year) and changes rarely. A static site has no running costs, no security surface and no moving parts to maintain.

GitHub Pages hosting instead of Firebase.
Since February 2026, Cloud Storage for Firebase requires the pay-as-you-go Blaze plan, which needs a payment card and has no hard spending cap. The organisation pays by EFT and has no card. GitHub Pages is free for public repositories and needs no payment details.

Content stored in Git.
Every change to results, documents or photos is a commit with an author and date, which gives a complete history and makes handover to a new maintainer straightforward.

Astro as the site generator.
Astro builds plain HTML pages from components and content files, and can generate pages directly from data such as club files and results tables.

3 Technology stack

Component	Choice
Site generator	Astro (exact version in `package.json`)
Runtime	Node.js 22.23.4 or newer (the build pipeline uses Node 24)
Package manager	npm (`package-lock.json` must stay committed)
Hosting	GitHub Pages
Build and deploy	GitHub Actions
Version control client	GitHub Desktop
Editor	Visual Studio Code

4 Repository and configuration

The repository is owned by the GitHub organisation `VTPSA`, not by an individual, so ownership survives a change of maintainer. It is public and named exactly `vtpsa.github.io`; this special name makes GitHub publish it at the root address with no path suffix.

Key files:

- `astro.config.mjs`: sets `site` to `https://vtpsa.github.io`. No `base` value is needed because the site is served from the root.
- `package.json` and `package-lock.json`: dependencies and their exact versions.
- `src/pages/`: each file here becomes a page of the site.
- `public/`: files copied to the site unchanged (for example PDFs and logos).
- `.github/workflows/deploy.yml`: the build and deploy pipeline.

5 Deployment pipeline

The workflow in `.github/workflows/deploy.yml` runs on every push to `main` and can also be started manually from the Actions tab of the repository. It has two jobs:

1. **build**: checks out the code, then runs the official `withastro/action`, which installs dependencies, builds the site and uploads the result. The Node version is set explicitly:

```
- name: Install, build, and upload your site
  uses: withastro/action@v3
  with:
    node-version: 24
```

2. **deploy**: publishes the built site using `actions/deploy-pages`.

In the repository settings, **Settings** → **Pages** → **Source** must be set to **GitHub Actions**. Publishing a change normally takes one to two minutes after the push.

6 Local development workflow

The project is kept outside OneDrive (for example in `C:\Projects`) because the `node_modules` folder contains thousands of small files that sync tools handle badly. The normal cycle is:

1. Open the project folder in Visual Studio Code.
2. Run the site locally from the project folder with `npm run dev` and view it at `http://localhost:4321`.
3. Edit files and check the result in the browser.
4. Commit and push using GitHub Desktop. The site then rebuilds and publishes automatically.

7 Problems encountered and their fixes

Problem	Cause and fix
Astro setup wizard looped in Git Bash	Interactive prompts do not work in Git Bash. Use PowerShell and pass the answers as command-line options.
`npm` blocked in PowerShell	Script execution is disabled by default on Windows. Fixed with `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned`.
Build failed: "Node.js v20 is not supported"	The Astro action defaults to Node 20 but this Astro version needs Node 22.23.4 or newer. Fixed by setting `node-version: 24` in the workflow.
"Site not found" at the root address	The repository was not named `vtpsa.github.io`. Renamed it and re-ran the workflow.

8 Notes for the next maintainer

- Keep the GitHub organisation owned by at least two people, and use an organisational contact email where possible.
- Do not delete `package-lock.json`; the build pipeline relies on it.
- If a build fails, open the repository's Actions tab, click the failed run, and read the log of the failing step.
- Version numbers in the workflow file (for example `withastro/action@v3`) may need updating over time; check the current Astro deployment guide.